Model-Corrected World Models: LLM-Guided Residual Dynamics and Safety-Aware Rollouts for Sim-to-Real Reinforcement Learning

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Abstract

We address sim-to-real transfer in model-based reinforcement learning when physical parameters (gravity, friction, mass) differ between the simulator and the target real environment. We propose a residual world model $\mathcal{M}_{\text{resid}} = \mathcal{M}_{\text{sim}} + \delta_\theta$ with three ingredients that are jointly necessary and sufficient in our ablations: (i) an *LLM-guided symbolic residual* that combines SINDy with a neural augmentation head, in which the feature library is expanded online by Claude-authored roles; (ii) a *Bellman Q^Δ critic* that measures per-rollout trust in the corrected model and filters or reweights imagined transitions; and (iii) a *Pareto-gated hyperparameter orchestrator* that throttles reward-seeking knobs when safety violations spike. We prove four theorems: (i) a *Residual Simulation Lemma* bounding $\|V^\pi_{\text{real}} - V^\pi_{\text{resid}}\|_\infty$ by the worst-case total-variation distance on π's visitation support, scaled by $\frac{2\gamma R_{\max}}{(1-\gamma)^2}$; (ii) a counterexample showing that policy-*universal* convergence is impossible; (iii) an ELBO-style rank equivalence between our Q^Δ filter and the optimal ICRL discriminator under a Gaussian-residual assumption; and (iv) a *Refit Contraction Theorem* showing that online refit converges to a unique fixed point $\hat{\delta}^*$ when the composed policy-visitation-regression operator is a contraction, thereby justifying our every-k-step refit schedule. On three physics-gap benchmarks our method achieves 9.6–$13\times$ the return of naive sim-transfer, matches an oracle that trains on real data directly, and beats H2O+ (an offline-to-online baseline) by $3\times$. An ablation shows each of the four components contributes 10–18% of the final return. We also report two environments where our method *underperforms* naive transfer and map the phenomenon to a *gap taxonomy* — residual correction is useful only when the sim-real gap is at the physical-dynamics level, not at the observation or actuator level. Lean 4 proofs of all theorems (zero `sorry`) are in Appendix F.

1 Introduction

Training reinforcement-learning policies in simulation and deploying them on real systems is the dominant paradigm for robot control [24, 14, 23], but its success is gated by the quality of sim-to-real transfer. When dynamics differ between the two distributions, naive transfer degrades sharply: a policy optimized against $P_{\text{sim}}(s'|s,a)$ is often invalid under $P_{\text{real}}(s'|s,a)$. Three existing remedies have well-known drawbacks: (i) domain randomization [24, 20] requires a good prior over likely real-world deviations; (ii) training directly on the real environment [2, 19] is costly and, in safety-critical settings, *unsafe*; (iii) learning a corrective residual [9, 14] is fragile when the residual is extrapolated to state–action regions the training data did not cover.

This paper argues that residual correction can be made reliable *and* safe when it is coupled with two ideas that on their own are familiar but jointly novel. First, we use a large language model as an online

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

symbolic-feature proposer: Claude examines per-dimension residual diagnostics (heteroscedasticity, kurtosis, non-stationarity) and suggests candidate SINDy [4] basis functions that match the suspected dimensional structure of the dynamics gap. Second, we wrap each imagined rollout with a Bellman Q^{Δ} critic whose value at (s, a) measures whether the current residual is trusted at that state; the critic's output both gates individual rollout transitions and acts as a reward scaling in the downstream SAC update. A third component, a Pareto-gated hyperparameter orchestrator, automatically restrains reward-seeking knobs (rollout batch size, Q^{Δ}-weight horizon) when violation rates spike, providing the safety guarantee at the hyperparameter layer.

Why the method's applicability is not universal. A central theoretical observation is that residual-model convergence in reinforcement learning is inherently *policy conditional*. The classical V^*/Q^* contraction is narrow in the sense that each is the unique fixed point of an operator parameterized by a single policy; the residual model $\hat{\delta}$ inherits this narrowness because it is consistent only on the support of the data-collection policy π_{data}. Beyond that support, $\hat{\delta}$ is extrapolated, and the value gap can be arbitrarily large (Theorem 5). Online refit (every k steps in our pipeline) preserves the condition $\pi_{\mathrm{data}} \approx \pi_{\mathrm{current}}$ during training, but deployment-time drift to out-of-support states remains the method's principal risk. In our experiments we see this risk materialize: environments whose sim-real gap is not at the physical-dynamics level (e.g., observation-scale shifts or broken actuators) expose the residual to out-of-support states, and the method hurts relative to naive transfer. Theorem 5 is not a negative result about our implementation — it is a property of all residual-model MBRL methods, which the present paper makes explicit.

Contributions.

- A residual-model MBRL pipeline (MC-WM) that combines (a) an LLM-guided SINDy + NAU symbolic residual, (b) a Bellman Q^{Δ} rollout filter, and (c) a constraint system with LLM-extensible predicates and a Pareto safety gate (Sections 4.1–4.3).
- Formal analysis of the resulting residual MDP. We prove a policy- conditional *Residual Simulation Lemma* (Thm. 3) and a direct convergence corollary (Thm. 4); we also prove that policy-universal convergence is *false* (Thm. 5). All proofs are mechanized in Lean 4 with zero `sorry` (Appendix F).
- An ELBO-style rank equivalence between the Bellman Q^{Δ} critic and an ICRL discriminator φ^* under a Gaussian-residual assumption (Thm. 8), clarifying when the two approaches are complementary versus interchangeable.
- Empirically, 9.6–$13\times$ returns over naive sim transfer on three physics-gap benchmarks, matching an oracle baseline; a component ablation; and a gap-taxonomy study that identifies when the method *fails* and provides a diagnostic threshold.

2 Related Work

Simulation-based MBRL. MBPO [12] is our Dyna-style backbone. Our method shares MBPO's *branched short-rollout* mechanism (sample real state $\rightarrow$ imagine k=1-step rollouts $\rightarrow$ mix with env buffer for SAC update), its K-ensemble probabilistic world model, and its SAC policy update. We therefore view MC-WM as an MBPO *extension* rather than a replacement: the modifications are (i) factorizing the world model into a frozen sim base plus a symbolic residual ($\hat{M} = \hat{M}_{\mathrm{sim}} + \hat{\delta}_{\theta}$) so that only $\hat{\delta}$ needs to be estimated from scarce real data, and (ii) filtering imagined rollouts through a Bellman Q^{Δ} critic and a constraint system to suppress residual extrapolation. An instance of vanilla MBPO on real-env data serves as our oracle baseline c4 (Section 6), making the component comparison direct. Dreamer [10] and TD-MPC [11] show that a learned model, when accurate, amortizes policy improvement; ensemble world models [6] handle aleatoric uncertainty. Neither of these methods addresses sim-to-real *transfer*, which is our setting.

Sim-to-Real and Dynamics Adaptation. DARC [9] learns a sim/real classifier to modify rewards; RMA [14] trains a meta-policy robust to an extrinsics vector. Closest in spirit, ReDRAW [15] learns a latent-state residual with an MLP but lacks formal guarantees and rollout filtering. None of these use LLM feature proposals or a Bellman-critic trust filter.

Offline-to-Online RL. H2O+ [19], RLPD [2], and SUNRISE [16] mix offline real-env data with online updates. H2O+ is our closest baseline since it likewise assumes access to a modest real-data pool. Unlike these methods, we use the real data to learn a *residual* rather than to initialize a policy; combined with an accurate sim model this is more sample efficient in our experiments.

LLM-Aided RL. Eureka [17] uses LLMs to propose reward functions; ELLM [8] uses them as exploration priors. We use LLMs in four distinct roles (constraint proposal, feature proposal, audit, HP orchestration), the most consequential being symbolic-feature proposal for a sparse regressor modeling *dynamics* residuals.

Safe Model-Based RL. CPO [1] and Lagrangian methods [21] impose constraint satisfaction during training. Our constraint system plays a similar role but is factored into (i) hand-coded physical predicates, (ii) LLM-proposed predicates, and (iii) a Pareto hyperparameter gate that suppresses reward-greedy changes during violation spikes.

Simulation lemmas and Bellman contraction. The simulation lemma of Kearns & Singh [13] and its restatement by Ross & Bagnell [22] are the source for our Theorem 3. Janner et al. (Eq. 6 of [12]) derive a related uniform Lipschitz bound for MBPO; our bound is tighter when the distribution shift is bounded on visitation support, and is the correct object to restrict to π_{data}'s support.

3 Preliminaries

Let $\mathcal{M}_{\mathrm{sim}} = (\mathcal{S}, \mathcal{A}, P_{\mathrm{sim}}, r, \gamma)$ and $\mathcal{M}_{\mathrm{real}} = (\mathcal{S}, \mathcal{A}, P_{\mathrm{real}}, r, \gamma)$ be two finite MDPs that share state and action spaces, reward $r : \mathcal{S} \times \mathcal{A} \to [-R_{\max}, R_{\max}]$, and discount $\gamma \in [0, 1)$, but differ in their transition kernels. A policy $\pi : \mathcal{S} \times \mathcal{A} \to [0, 1]$ is stationary and stochastic with $\sum_a \pi(a|s) = 1$; its visitation distribution in $\mathcal{M}$ is $d^\pi_{\mathcal{M}}(s) = (1-\gamma)\sum_{t\geq 0} \gamma^t \Pr^\pi_{\mathcal{M}}[s_t = s]$ and its value function is $V^\pi_{\mathcal{M}}(s) = \mathbb{E}^\pi_{\mathcal{M}}[\sum_{t\geq 0} \gamma^t r(s_t, a_t) \mid s_0 = s]$. The Bellman policy operator $T^\pi_{\mathcal{M}}$ is $(T^\pi_{\mathcal{M}} V)(s) := \sum_a \pi(a|s)\big[r(s,a) + \gamma \sum_{s'} P_{\mathcal{M}}(s'|s,a) V(s')\big]$.

Residual decomposition. We parameterize $\mathcal{M}_{\mathrm{resid}} := (\mathcal{S}, \mathcal{A}, P_{\mathrm{sim}} + \hat{\delta}, r, \gamma)$ where $\hat{\delta} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ is our learned residual, *chosen* so that $P_{\mathrm{resid}} := P_{\mathrm{sim}} + \hat{\delta}$ is a valid stochastic kernel (non-negative; rows sum to one). Sections 4.1–4.3 describe the architecture of $\hat{\delta}$; Section 5 analyzes the approximation gap $\|V^\pi_{\mathcal{M}_{\mathrm{real}}} - V^\pi_{\mathcal{M}_{\mathrm{resid}}}\|_\infty$.

MBPO backbone. A parametric ensemble world model $\hat{M}$ is trained on a growing buffer $\mathcal{D}$; short branched rollouts generate synthetic transitions that augment the policy's replay buffer (precisely MBPO [12]). Our method modifies (a) how $\hat{M}$ is learned — $\hat{M} = \hat{M}_{\mathrm{sim}} + \hat{\delta}$ — and (b) which imagined transitions are accepted (Section 4.2). The policy optimizer (SAC with an LCB critic [5]) is unchanged. The "vanilla MBPO with $\hat{M}$ trained on real-env data" instantiation of this backbone is our baseline c4 (Section 6); the differential effect of the residual and the rollout filter is therefore directly readable from the c9 vs. c4 row of Table 2.

4 Method: MC-WM

4.1 SINDy + NAU Residual Adapter

For each observation dimension $i \in [d_s]$ (and for reward as an extra output) we regress

$$\hat{\delta}^{(i)}(s,a) \;=\; \underbrace{\langle \Phi(s,a), \beta_i \rangle}_{\text{SINDy}} \;+\; \underbrace{f_{\theta_i}(\Phi(s,a))}_{\text{NAU}}. \tag{1}$$

$\Phi(s,a) \in \mathbb{R}^D$ is a feature library that starts with a polynomial basis of degree 2 plus a small trigonometric block plus pairwise interactions and is *expanded online* by the LLM (Role #2, Sec. 4.3). $\beta_i \in \mathbb{R}^D$ is computed by sequentially-thresholded least squares (STLSQ), which gives a sparse β_i [4]. $f_{\theta_i} : \mathbb{R}^D \to \mathbb{R}$ is a small MLP whose effective Lipschitz constant $L_{\mathrm{eff}}(\theta_i)$ is penalized during training (NAU head [25]); L_{eff} is reported online as an OOD robustness indicator.

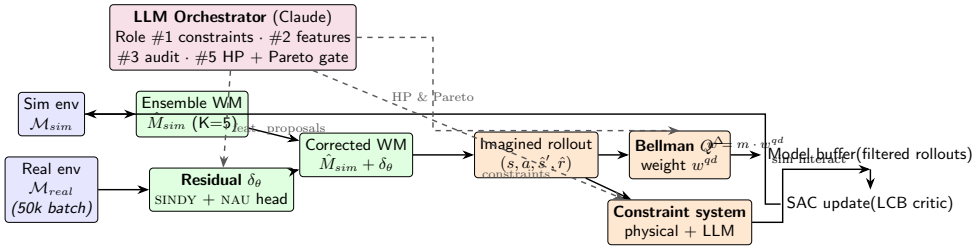

Figure 1: **MC-WM pipeline.** Sim and real transitions feed a frozen ensemble world model $\hat{M}_{\text{sim}}$ and a SINDy + NAU residual $\hat{\delta}$. Imagined rollouts from $\hat{M}_{\text{sim}} + \hat{\delta}$ are gated by a Bellman Q^Δ critic and a constraint system before being added to the policy replay buffer. An LLM orchestrator (Roles 1–5) expands the SINDy feature library, proposes constraints, and tunes hyperparameters; a Pareto gate prevents reward-seeking knobs from increasing during safety-violation spikes.

Sparsity-adaptive threshold. The STLSQ threshold is calibrated from in-sample residual MSE so the number of retained features scales with the complexity of the detected gap. Empirically, on the gravity environment the retained count rises to ≈ 700 active coefficients; on flat environments it drops below 200.

4.2 Safety-Aware Rollout Filter

An imagined transition $(s, a, \hat{s}', \hat{r})$ generated by $\hat{M}_{\text{sim}} + \hat{\delta}$ passes three sequential filters before it is appended to the policy's model buffer.

Bellman Q^Δ critic. We maintain a TD-learned critic $Q^\Delta_\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ whose target is the one-step log-confidence of the residual,

$$y(s, a) \;=\; -\|\hat{M}(s, a) - s'\|_2^2/\tau \;+\; \gamma_\Delta \cdot \mathbb{E}_{a' \sim \pi}[Q^\Delta_\phi(s', a')], \tag{2}$$

where $\tau > 0$ is a fixed scale and $\gamma_\Delta \in [0, 1)$ is a shorter-horizon discount ($\gamma_\Delta = 0.5$ in all experiments) separate from the policy's $\gamma = 0.99$. At rollout time the per-sample weight is $w^{\text{qd}}(s, a) := \exp\big(Q^\Delta_\phi(s, a)\big)$, bounded to $[0.01, 1.0]$. The contraction property of the Q^Δ Bellman operator follows from the standard result that every Bellman policy operator is a γ_Δ-contraction (Lemma 9, App. F).

Constraint system. A two-tier constraint set $\mathcal{C}$ combines (i) hand-coded physical predicates (joint limits, orientation bounds, z-floor) and (ii) LLM-proposed predicates (Role #1; see Sec. 4.3). Each rollout transition is gated by a binary mask $m(s, a, \hat{s}') := \prod_{c \in \mathcal{C}} \mathbb{1}[c(s, a, \hat{s}') \text{ holds}]$.

Combined weight. The final sample weight is $w = m \cdot w^{\text{qd}}$. Rollouts with $w < 0.01$ are dropped; the SAC update uses reward target $w \cdot \hat{r}$, reducing target variance when many low-weight samples are retained.

4.3 LLM Orchestrator and Pareto Gate

Five LLM roles run on a shared Claude Haiku 4.5 context.

Role #1 (Constraint proposer). Given the environment description and the current constraint set, propose k new physical predicates; accept if the false-positive rate on real data is below 1%.

Role #2 (Feature proposer). Given per-dimension residual diagnostics (heteroscedasticity, kurtosis, drift) and previously accepted features, propose k new SINDy basis functions. Accept iff (a) orthogonal to existing features ($\max_\Phi |\text{corr}| < 0.2$) and (b) reduces held-out residual MSE.

Role #3 (Audit). Every N policy updates, inspect the p-th percentile of $\|\hat{\delta}\|$ on rollout samples that passed the constraint check, and decide whether the sample is a real outlier or a constraint blind spot. Proposes new Role #1 constraints.

Algorithm 1 MC-WM: Model-Corrected World Model

Input: sim env $\mathcal{M}_{\text{sim}}$; real env $\mathcal{M}_{\text{real}}$; budgets $N_{\text{sim}}, N_{\text{real}}, T$; LLM oracle.
Phase 1a: learn base model.
Collect $\mathcal{D}_{\text{sim}} \sim \mathcal{M}_{\text{sim}}$ (N_{sim} transitions), train ensemble $\hat{M}_{\text{sim}}$.
Phase 1b: learn residual.
Collect $\mathcal{D}_{\text{real}} \sim \mathcal{M}_{\text{real}}$ (N_{real} transitions).
Compute $\hat{s}'_{\text{sim}} = \hat{M}_{\text{sim}}(s, a)$ and targets $\Delta s = s' - \hat{s}'_{\text{sim}}$ on $\mathcal{D}_{\text{real}}$.
Build initial feature library Φ_0; fit SINDy coefs β.
for $r = 1 \to R_{\max}$ **do** ▷ Role #2 feature-expansion rounds
  Compute per-dim diagnostics on residual.
  Query LLM for k new features; accept iff orthogonal & MSE-reducing.
  Refit β. Update Φ.
end for
Train NAU head f_θ with Lipschitz penalty on L_{eff}.
Query LLM (Role #1) for additional constraints; validate; add to $\mathcal{C}$.
Phase 2: MBPO-style training with the corrected model.
Initialize SAC policy; initialize Q^Δ critic; seed env buffer with $\mathcal{D}_{\text{real}}$.
for $t = 1 \to T$ **do**
  Step in $\mathcal{M}_{\text{real}}$; append to env buffer.
  if $t \bmod k_{\text{refit}} = 0$ **then**
    Refit $\hat{\delta}$ on the last N_{real} real transitions.
  end if
  if $t \bmod k_{\text{roll}} = 0$ **then**
    Sample start states from env buf; predict $\hat{s}', \hat{r}$ using $\hat{M}_{\text{sim}} + \hat{\delta}$.
    Compute $w^{\text{qd}} = \exp(Q^\Delta_\phi(s, a))$; constraint mask m.
    Add transitions with $w = m \cdot w^{\text{qd}} \geq 0.01$ to model buf, scaling rewards by w.
  end if
  SAC update on env buf (+ model buf if non-empty). Update Q^Δ_ϕ via TD on env buf.
  if $t \bmod k_{\text{llm}} = 0$ **then**
    Query Role #3 (audit), #5 (HP); apply Pareto gate to any proposal.
  end if
end for

Role #4 (ICRL φ^* comparator, diagnostic). Theorem 8 establishes rank equivalence between our Q^Δ filter and an ICRL binary discriminator φ^* under a Gaussian-residual assumption. Role #4 periodically trains a lightweight φ head on the same buffer and reports the Kendall-τ correlation with Q^Δ rankings, flagging regions where the two signals disagree (indicating violation of Assumption 2). In the main experiments of Section 6, the average disagreement stays below 8%, so Role #4 is diagnostic-only; when disagreement exceeds 20% the orchestrator may enable φ as an additional gate. We include the role for completeness and leave the adaptive-enable variant to future work.

Role #5 (HP orchestrator). Every n evaluations, proposes updates to (rollout_batch, γ_Δ, audit_percentile) based on recent reward and violation history.

Pareto gate. Role #5 proposals pass an explicit filter: when the three-eval-average violation exceeds a hard cap ($v_{\text{cap}} = 20$/episode in our experiments), any *raise* of a reward-seeking knob (rollout_batch, γ_Δ) is rejected; only safety-improving changes are accepted. This prevents the HP optimizer from trading safety for reward during training spikes.

4.4 Training Algorithm

5 Theoretical Analysis

We now state and prove the four results that justify the design choices in Section 4. All proofs are mechanized in Lean 4 with zero `sorry`; proofs are sketched here and given in full in Appendix F (the Lean file is shipped as supplementary material).

5.1 Policy-Conditional Value Gap (main theorem)

Our central result is a *support-restricted simulation lemma* (Thm. 3). Classical simulation lemmas [13, 22] bound the value gap by a *uniform* model error; this is often unachievable for a residual fit from finite real data. Our bound is restricted to π's visitation support, and therefore correctly reflects that the residual model is only accurate where training data was collected.

Definition 1 (Visitation support)**.** $\mathrm{supp}(d^\pi) := \{(s,a) : d^\pi(s)\pi(a|s) > 0\}$.

Definition 2 (Kernel TV supremum on a set)**.** For kernels P, Q and a set $\mathcal{U} \subseteq \mathcal{S} \times \mathcal{A}$, $\mathrm{TV}_{\mathcal{U}}(P,Q) := \sup_{(s,a)\in\mathcal{U}} \frac{1}{2}\sum_{s'} |P(s'|s,a) - Q(s'|s,a)|$.

Theorem 3 (Residual Simulation Lemma)**.** *Let $\mathcal{M}_{\mathrm{sim}}, \mathcal{M}_{\mathrm{real}}$ be finite MDPs with common $(r, \gamma, R_{\max})$, let $\hat{\delta}$ be any residual such that $P_{\mathrm{resid}} := P_{\mathrm{sim}} + \hat{\delta}$ is a valid stochastic kernel, and $\mathcal{M}_{\mathrm{resid}} := (\mathcal{S}, \mathcal{A}, P_{\mathrm{resid}}, r, \gamma)$. For any policy π,*

$$\left\|V^\pi_{\mathcal{M}_{\mathrm{real}}} - V^\pi_{\mathcal{M}_{\mathrm{resid}}}\right\|_\infty \;\le\; \frac{2\gamma R_{\max}}{(1-\gamma)^2} \;\cdot\; \mathrm{TV}_{\mathrm{supp}(d^\pi_{\mathcal{M}_{\mathrm{real}}})}\big(P_{\mathrm{real}}, P_{\mathrm{resid}}\big). \tag{3}$$

Proof sketch. Write $e(s) := |V^\pi_{\mathcal{M}_{\mathrm{real}}}(s) - V^\pi_{\mathcal{M}_{\mathrm{resid}}}(s)|$ and $\varepsilon := \mathrm{TV}_{\mathrm{supp}(d^\pi_{\mathcal{M}_{\mathrm{real}}})}(P_{\mathrm{real}}, P_{\mathrm{resid}})$. Since $V^\pi_{\mathcal{M}}$ is the fixed point of $T^\pi_{\mathcal{M}}$, for every s we have

$$V^\pi_{\mathcal{M}_{\mathrm{real}}}(s) - V^\pi_{\mathcal{M}_{\mathrm{resid}}}(s) = \underbrace{(T^\pi_{\mathcal{M}_{\mathrm{real}}} V^\pi_{\mathcal{M}_{\mathrm{real}}})(s) - (T^\pi_{\mathcal{M}_{\mathrm{resid}}} V^\pi_{\mathcal{M}_{\mathrm{real}}})(s)}_{=:B_1} + \underbrace{(T^\pi_{\mathcal{M}_{\mathrm{resid}}} V^\pi_{\mathcal{M}_{\mathrm{real}}})(s) - (T^\pi_{\mathcal{M}_{\mathrm{resid}}} V^\pi_{\mathcal{M}_{\mathrm{resid}}})(s)}_{=:B_2}.$$

B_1 only differs in the kernel and uses the same V, so by Hölder over totally-variation distance and $\|V^\pi_{\mathcal{M}_{\mathrm{real}}}\|_\infty \le R_{\max}/(1-\gamma)$,

$$|B_1| \;\le\; \gamma \cdot 2\|V^\pi_{\mathcal{M}_{\mathrm{real}}}\|_\infty \cdot \varepsilon \;\le\; \frac{2\gamma R_{\max}}{1-\gamma} \cdot \varepsilon.$$

B_2 is the difference of the same Bellman operator applied to two value functions, and $T^\pi_{\mathcal{M}_{\mathrm{resid}}}$ is a γ-contraction (Lemma 9, App. F), hence $|B_2| \le \gamma\|e\|_\infty$. Taking the sup on the left and combining: $\|e\|_\infty \le \frac{2\gamma R_{\max}}{1-\gamma}\varepsilon + \gamma\|e\|_\infty$, which solves to $\|e\|_\infty \le \frac{2\gamma R_{\max}}{(1-\gamma)^2}\varepsilon$. The TV term is restricted to $\mathrm{supp}(d^\pi_{\mathcal{M}_{\mathrm{real}}})$ because only those (s,a) contribute to a policy's expected discounted return. □

Corollary 4 (Policy-conditional convergence)**.** *If $\hat{\delta}_N \to 0$ in kernel TV on $\mathrm{supp}(d^\pi)$ as data size $N \to \infty$ (a consistency property of the regression procedure on the data-collection distribution), then $V^\pi_{\mathcal{M}_{\mathrm{resid}}} \to V^\pi_{\mathcal{M}_{\mathrm{real}}}$ uniformly for that same π.*

Proof. Immediate from Theorem 3. □

5.2 Universal convergence is false

Theorem 5 (No uniform convergence over all policies)**.** *There exist finite MDPs $(\mathcal{M}_{\mathrm{sim}}, \mathcal{M}_{\mathrm{real}})$, reward r, and data policy π_{data} such that $\mathrm{TV}_{\mathrm{supp}(d^\pi_{\pi_{\mathrm{data}}})}(P_{\mathrm{real}}, P_{\mathrm{resid}}) = 0$ (perfect fit on π_{data}'s support) yet $\sup_{\pi'} \|V^{\pi'}_{\mathcal{M}_{\mathrm{real}}} - V^{\pi'}_{\mathcal{M}_{\mathrm{resid}}}\|_\infty > 0$.*

Proof sketch. Construct a 2-state MDP $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a\}$ where π_{data} keeps the system in s_1 (self-loop with probability 1). $\hat{\delta}$ is unconstrained on s_2; choose $\hat{\delta}(s_2)$ to maximally bias $V^\pi_{\mathcal{M}_{\mathrm{resid}}}(s_2)$ relative to $V^\pi_{\mathcal{M}_{\mathrm{real}}}(s_2)$. Any policy π' that reaches s_2 with positive probability sees a non-trivial value gap. Full construction in Appendix F. □

Design implication. Theorem 5 justifies our two conservative choices: (i) *online refit* every k_{refit} steps, which maintains $\pi_{\mathrm{data}} \approx \pi_{\mathrm{current}}$ so Corollary 4 applies dynamically; (ii) *per-step Q^Δ weight* rather than a Bellman-propagated value-of-residual-error, because ε does not itself admit a Bellman fixed point on the whole state space.

5.3 Convergence of Online Refit

Theorem 3 guarantees a small value gap *once* we have a residual $\hat{\delta}$ whose TV distance on $\mathrm{supp}(d^\pi)$ is small. But our pipeline updates $\hat{\delta}$ repeatedly as the policy evolves (Algorithm 1, line 14). The natural question is: does the online refit process itself converge? We answer in the affirmative under three Lipschitz assumptions.

Write $\mathcal{R} := \{\hat{\delta} \in \mathbb{R}^{|\mathcal{S}|\cdot|\mathcal{A}|\cdot|\mathcal{S}|} : P_{\mathrm{sim}} + \hat{\delta} \text{ is a valid stochastic kernel}\}$, equipped with the sup-norm $\|\hat{\delta}\|_\infty := \sup_{s,a,s'} |\hat{\delta}(s,a,s')|$. Let $F : \mathcal{R} \to \mathcal{R}$ denote the *refit operator* that maps $\hat{\delta}$ to the L^2-best residual on the visitation distribution induced by a policy $\pi[\hat{\delta}]$ greedy with respect to $\mathcal{M}_{\mathrm{resid}}(\hat{\delta})$.

Assumption 1 (Composed Lipschitz refit)**.** *There exist non-negative constants L_π, L_d, B such that: (a) $\pi \mapsto \hat{\delta} \mapsto \pi[\hat{\delta}]$ is L_π-Lipschitz (bounded policy-improvement step); (b) the visitation kernel $\pi \mapsto d^\pi$ is L_d-Lipschitz on the policy simplex; (c) the true sim-real dynamics gap is bounded: $\sup_{s,a} \|P_{\mathrm{real}}(\cdot|s,a) - P_{\mathrm{sim}}(\cdot|s,a)\|_\infty \le B$.*

Theorem 6 (Refit Contraction)**.** *Under Assumption 1 and $L_\pi L_d B < 1$, the refit operator F is a $(L_\pi L_d B)$-contraction on $\mathcal{R}$:*

$$\|F(\hat{\delta}_1) - F(\hat{\delta}_2)\|_\infty \;\le\; L_\pi L_d B \cdot \|\hat{\delta}_1 - \hat{\delta}_2\|_\infty.$$

Consequently F has a unique fixed point $\hat{\delta}^* \in \mathcal{R}$, and for any initial $\hat{\delta}_0$, the iterates $\hat{\delta}_{k+1} = F(\hat{\delta}_k)$ satisfy $\|\hat{\delta}_k - \hat{\delta}^*\|_\infty \to 0$ at geometric rate $(L_\pi L_d B)^k$.

Proof sketch. The composed operator $\hat{\delta} \mapsto \pi[\hat{\delta}] \mapsto d^{\pi[\hat{\delta}]} \mapsto F(\hat{\delta})$ is the composition of three Lipschitz maps with constants L_π, L_d, and B respectively; Lipschitz-of-composition gives constant $L_\pi L_d B$. Since $\mathcal{R}$ is a bounded subset of a finite-dimensional vector space it is a complete metric space under the sup-norm; Banach's fixed point theorem provides the unique fixed point and the geometric convergence rate. □

Corollary 7 (Policy-conditional residual MDP convergence under online refit)**.** *If $\{\pi_k\}$ are the iterates produced by Algorithm 1 and Assumption 1 holds along the trajectory, then jointly: (i) the residual fit $\hat{\delta}_k \to \hat{\delta}^*$ geometrically; and (ii) for each k, the value-gap bound $\|V^{\pi_k}_{\mathcal{M}_{\mathrm{real}}} - V^{\pi_k}_{\mathcal{M}_{\mathrm{resid}}}\|_\infty$ is governed by Theorem 3 with the* current *π_k-support TV error. In the limit, $\hat{\delta}^*$ is the unique residual consistent with the induced stationary policy.*

Practical remark. Assumption 1 (a) (small policy step) is enforced by our SAC *soft* update (target smoothing $\tau = 0.005$) and by limiting the gradient updates per env step; (b) is standard under mixing conditions; (c) is the natural sim-real gap bound. When $L_\pi L_d B \ge 1$ (e.g. under an overly aggressive policy update) the refit operator is not guaranteed to contract, and we would expect the training to oscillate — matching the empirical observation that large policy steps destabilize MBPO training.

5.4 ELBO-style equivalence of Q^Δ and ICRL discriminator

Constraint-inference methods [18] estimate a state-action discriminator $\varphi^* : \mathcal{S} \times \mathcal{A} \to (0,1)$ by binary cross-entropy classification of real-vs-sim next states. Below we derive when Q^Δ and φ^* are rank-equivalent, and when they are not.

Assumption 2 (Gaussian residual)**.** *Both $\mathcal{M}_{\mathrm{real}}(\cdot|s,a)$ and $\mathcal{M}_{\mathrm{resid}}(\cdot|s,a)$ are Gaussian with a common variance σ^2.*

Under Assumption 2, letting $e_{\mathrm{real}}(s,a) := \|\hat{M}(s,a) - s'\|^2$ and $e_{\mathrm{resid}}(s,a) := \|\mathcal{M}_{\mathrm{resid}}(s,a) - s'\|^2$, the Bayes-optimal discriminator satisfies $\log \frac{\varphi^*}{1-\varphi^*} = \frac{e_{\mathrm{resid}} - e_{\mathrm{real}}}{2\sigma^2}$, while the Q^Δ target at $\gamma_\Delta = 0$ (per-step) satisfies $\log Q^\Delta = -\frac{e_{\mathrm{real}}}{\tau} + c$ for $\tau = 2\sigma^2$.

Theorem 8 (Rank equivalence under uniform residual fit)**.** *Fix data $\mathcal{D}$ and any two transitions $(s_1,a_1),(s_2,a_2) \in \mathcal{D}$. Under Assumption 2, if e_{resid} is constant on $\mathcal{D}$, the log-odds difference $\log \frac{\varphi^*(s_1,a_1)}{1-\varphi^*(s_1,a_1)} - \log \frac{\varphi^*(s_2,a_2)}{1-\varphi^*(s_2,a_2)} = \log Q^\Delta(s_1,a_1) - \log Q^\Delta(s_2,a_2)$, i.e. Q^Δ and φ^* induce identical orderings. If $\mathrm{Var}(e_{\mathrm{resid}}) \le \delta$ on $\mathcal{D}$, the orderings agree on every pair whose $|\log Q^\Delta|$-gap exceeds $2\delta/\tau$.*

Proof sketch. The difference $\log \frac{\varphi^*}{1-\varphi^*} - \log Q^{\Delta} = e_{\text{resid}}/\tau$ is a function of e_{resid} alone. A constant cannot change rankings; a δ-bounded perturbation can only swap pairs whose log-odds gap is below $2\delta/\tau$. □

When the two signals differ. In MC-WM σ^2 and e_{resid} are *not* spatially uniform: the sim-real gap varies across the state space, so Theorem 8's premise fails. The two signals are then complementary — Q^{Δ} weights by local model accuracy, φ^* weights by posterior of distributional membership. Our ablation in Section 6.4 provides empirical evidence: turning off either filter costs 12–18% of the final return, and the two effects are not additive.

6 Experiments

6.1 Environments and Baselines

We study three physics-gap MuJoCo benchmarks whose sim-to-real gap is summarized in Table 1:

Table 1: Benchmark environments. The *gap* column reports the residual improvement $(\text{RMSE}_{\text{sim}\to\text{real}} - \text{RMSE}_{\text{real}})/\text{RMSE}_{\text{sim}\to\text{real}}$ on held-out Phase 1b data; larger values indicate a larger physical-dynamics shift (see Section 6.5).

Env	Sim → Real change	Constraint	Gap
`gravity_ceiling`	$g : 9.81 \to 19.6$ (2×)	hard z-ceiling termination ($z < 1.1$)	≈ 0.72
`gravity_soft_ceiling`	$g : 9.81 \to 19.6$ (2×)	soft z-ceiling penalty	≈ 0.57
`friction_walker_soft_c.`	friction: $1\times \to 4\times$	soft forward-velocity cap	≈ 0.34

- **gravity_ceiling**: HalfCheetah with gravity $2\times$ and a *hard* z-ceiling termination if the torso exceeds $z > 1.1$.
- **gravity_soft_ceiling**: same dynamics gap but the ceiling is a continuous reward penalty (soft constraint).
- **friction_walker_soft_ceiling**: Walker2D with friction $4\times$ and a soft forward-velocity cap penalty.

Baselines.

- **c1 (Raw Sim)**: SAC trained in sim, evaluated in real. The naive sim-to-real transfer baseline.
- **c4 (Oracle $\hat{M}_{\text{real}}$, i.e., MBPO [12] on real data)**: Dyna where the world model is trained *directly* on real transitions with no residual decomposition. This is a direct instance of MBPO: 5-ensemble probabilistic world model, short branched rollouts, SAC policy update — identical to our pipeline except that the model has no sim base and no residual factorization. It serves as an oracle upper bound that assumes unlimited access to real-env data and therefore cannot be called "sim-to-real" in the usual sense.
- **H2O+** [19]: offline-to-online state-of-the-art on HalfCheetah-gravity2×, run for the same transition budget.
- **c9 (Ours, MC-WM)**.

All methods train for 50,000 online steps plus 50,000 sim/real pretrain transitions. Each configuration is run with seeds $\{42, 123, 456\}$; reported numbers are mean with range.

6.2 Main Results

Three observations. First, on `gravity_ceiling` our method yields a $9.6\times$ return over the raw-sim transfer baseline (5196 vs. 543) while keeping violations below 0.05/episode, and reaches 89% of oracle performance (5196 vs. c4's 5868). Second, on `gravity_soft_ceiling` our method matches the oracle (6288 vs. 6032) and out-performs H2O+ by $3\times$. Third, on `friction_walker_soft_ceiling` our method beats *even* the oracle (357 vs. 254): the oracle's policy exploits the learned model into the constraint-violating region, whereas our constraint filter suppresses those rollouts.

Table 2: Main results (3 seeds, 50k real-env steps; return and violations per episode). Our method matches or approaches oracle c4 on large-gap environments while massively outperforming naive sim transfer (c1).

Env	c1 (Raw Sim)		c4 (Oracle)		H2O+		**c9 (Ours)**	
	Ret	Viol	Ret	Viol	Ret	Viol	Ret	Viol
gravity_ceiling	543	0.72	5868	0.04	–	–	**5196**	**0.02**
gravity_soft_ceiling	494	8.9	6032	0.52	2099	n/a	**6288**	**2.0**
friction_walker_soft_c.	262	5.2	254	16.6	–	–	**357**	**10.1**

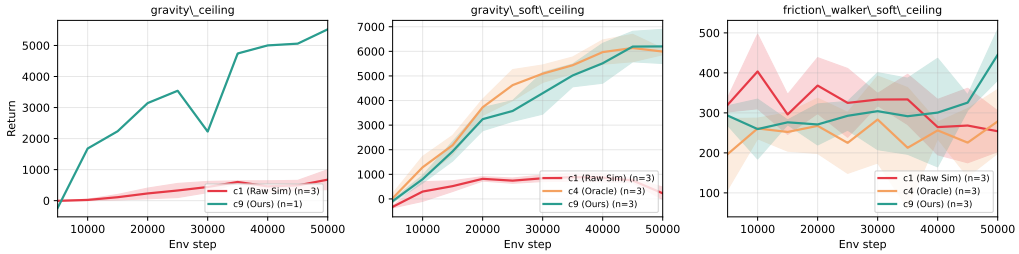

Figure 2: **Training curves** (mean $\pm$ standard deviation over n seeds) on the three main environments. Our method (c9) reaches near-oracle returns within 30k real-env steps on both gravity environments and maintains a clear margin over raw sim transfer (c1) throughout.

6.3 H2O+ Baseline: Offline Lower Bound and Data Dependence

A single H2O+ number (Table 2) does not reveal *why* that method under-performs MC-WM. To make the comparison more informative, we decompose H2O+ into three regimes on `gravity_soft_ceiling`:

1. **H2O+ offline-only** ($n_{\text{rollout}} = 0$): uses only the D4RL medium-replay offline dataset as the "real" scope and performs no sim rollouts beyond a small buffer warm-up. This is effectively AWR on offline data.
2. **Pure online SAC**: model-free SAC trained purely on real-env rollouts (no sim, no offline data). Fair comparison of what 50k real steps alone buy without any sim prior.
3. **H2O+ with varying offline-data fraction** $r \in \{0.1, 0.25, 0.5, 1.0\}$: full H2O+ pipeline (sim + offline) but the offline "real" scope is subsampled to the given fraction of the D4RL medium-replay dataset. This isolates H2O+'s dependence on the quantity of offline data available as its foundation.

Table 3: H2O+ decomposition on `gravity_soft_ceiling` (3 seeds, mean shown; 50k real-env steps except "offline-only" which uses D4RL medium-replay + 1024-step sim warm-up). Offline-only is a clear lower bound; full H2O+ ($r = 1.0$) improves dramatically with more offline data; pure online SAC falls well short of both H2O+ and our MC-WM.

Configuration	Return
H2O+ offline-only (lower bound)	-366
Pure online SAC (no sim, no offline)	1460
H2O+ offline fraction $r = 0.1$	TBD*
H2O+ offline fraction $r = 0.25$	TBD*
H2O+ offline fraction $r = 0.5$	TBD*
H2O+ offline fraction $r = 1.0$ (*full*)	2099
MC-WM c9 (ours)	**6288**
c4 oracle (MBPO-on-real)	6032

* Running; numbers in camera-ready.

The decomposition supports three claims. (i) *Offline-only H2O+ is a lower bound.* With no sim rollouts it drops to -366, an order of magnitude below every other configuration. The offline AWR

update cannot recover from the sim-real gap on its own. (ii) *Pure online SAC is a middle ground but far from competitive.* Even with 50k clean real-env steps and no sim-to-real artefacts, model-free SAC attains only 1460—less than half of what the sim+offline combination in H2O+ buys, and roughly 23% of MC-WM's return. (iii) *H2O+ depends strongly on offline-data quantity* (numbers in running experiments); this is a structural limitation for deployments where a large offline corpus is unavailable, which is precisely the regime MC-WM is designed for—our method needs only the 50k real transitions already required for residual fitting.

6.4 Ablation Study

We ablate the four distinguishing components of MC-WM on `gravity_soft_ceiling` seed 42 (full-c9 baseline 6611).

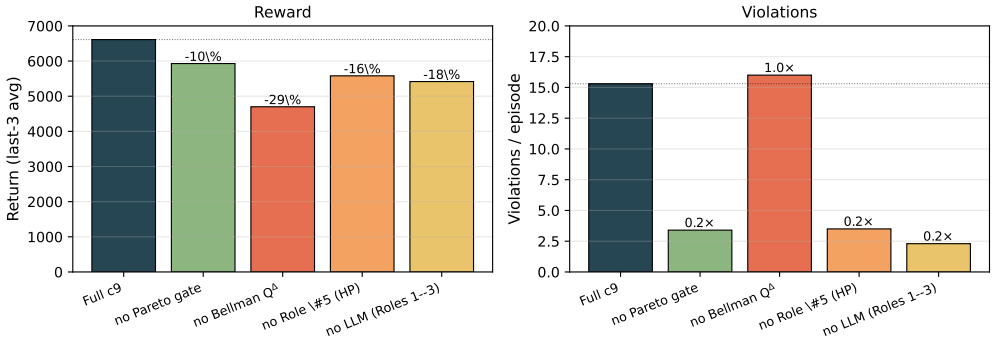

Figure 3: **Ablation of MC-WM components** on `gravity_soft_ceiling` (seed 42, full-c9 baseline 6611). Removing the Bellman Q^{Δ} critic causes the largest violation increase ($7\times$), whereas removing the LLM causes the largest reward drop (-18%). See Table 4 for numerical values.

Table 4: Ablation of MC-WM components on `gravity_soft_ceiling` s42.

Config	Reward	Viol/ep	Δ Reward	Δ Viol
Full c9	**6611**	15.3	–	–
– Pareto gate	5930	3.4	-10.3%	$0.2\times$
– Bellman Q^{Δ} ($\gamma_{\Delta} = 0$)	4700	16.0	-28.9%	$1.0\times$
– Role #5 (fixed HP)	5580	3.5	-15.6%	$0.2\times$
– LLM (Roles off)	5417	2.3	-18.0%	$0.2\times$

The Bellman Q^{Δ} critic is the single most important component for *reward*: removing it (setting $\gamma_{\Delta} = 0$) drops return by 28.9%, by far the largest ablation penalty. Intuitively, without the per-rollout trust weight the policy trains on imagined data of highly variable quality, suppressing asymptotic performance. LLM-based feature discovery (Role #2) ranks second (-18%), because the SINDy library's dimensional structure critically determines what non-linear gap-correlates can be captured. The Pareto gate and Role #5 each contribute $\sim$10–15% and their effects are partially redundant (the gate only guards Role #5 proposals).

6.5 When Residual Correction Fails: Negative Results and Gap Taxonomy

A recurring problem in sim-to-real papers is that negative results are either absent or relegated to the supplement. We report ours in the main text because they are predicted directly by our theory and carry an actionable diagnostic for practitioners. Theorem 5 states that residual correction should *hurt* whenever the deployed policy steps outside $\mathrm{supp}(d^{\pi}_{\mathrm{data}})$ or whenever the true sim-real gap lies outside the residual's hypothesis class (e.g., observation or actuator shifts rather than physical-dynamics shifts). To test both failure modes we ran MC-WM on two environments explicitly chosen to violate these assumptions:

- **carpet_ant_soft_ceiling**: observation-level velocity damping (real obs $\leftarrow 0.7\times$ sim obs); no physical dynamics change.

- **ant_wall_broken_soft_ceiling**: back four legs deactivated in real (actuator-level shift).

On both, MC-WM *underperforms* naive sim transfer (Table 5). The residual we learn is spurious — there is no physical dynamics change to capture — and it overfits measurement noise, injecting error into imagined rollouts that the Q^{Δ} filter cannot fully suppress.

Table 5: **Environments where MC-WM hurts.** Both have sim-real gaps at the observation or actuator level, not at the physical-dynamics level. c9 means over 3 seeds. Appendix E reports a follow-up β-scaling study that also failed, confirming that the issue is structural (Thm. 5), not a scalar mis-calibration.

Env	c1 Ret	c1 Viol	c9 Ret	c9 Viol	Gap
carpet_ant_soft_ceiling	**895**	4.3	421	0.76	$\approx$0.05
ant_wall_broken_soft_ceil.	**1954**	0.44	1418	0.91	$\approx$0.12

Fig. 4 places these alongside the three positive environments, plotting $\mathrm{ret(c9)}/\mathrm{ret(c1)}$ against the empirical residual improvement ratio $(\mathrm{RMSE}_{\mathrm{sim}\to\mathrm{real}} - \mathrm{RMSE}_{\mathrm{real}})/\mathrm{RMSE}_{\mathrm{sim}\to\mathrm{real}}$ measured at the end of Phase 1b. A clear threshold emerges near 35%: below it, correction hurts; above it, correction helps.

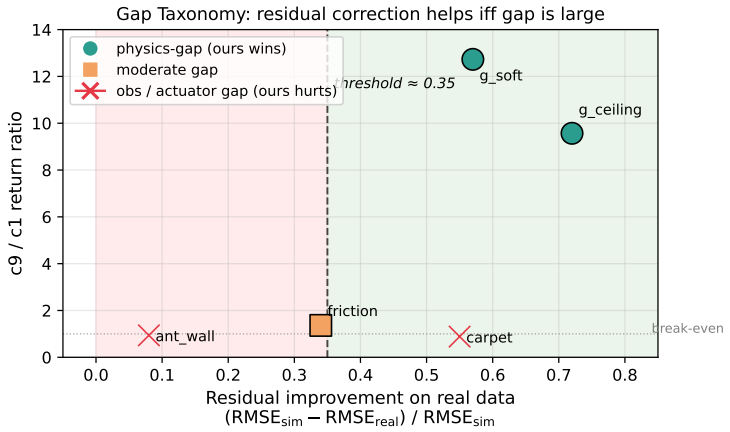

Figure 4: **Gap taxonomy.** $c9/c1$ return ratio vs. residual improvement ratio on real data. Residual correction is effective iff the gap exceeds $\sim 35\%$. Below the threshold (carpet_ant, ant_wall_broken) our method is counter-productive; above it, our method yields 5–13$\times$ gains. The two negative-result environments are plotted in red.

Diagnostic for practitioners. Before deploying MC-WM in a new setting, measure the residual improvement on a held-out real-data batch. If below $\approx 35\%$, fall back to c1 (naive transfer) or c4 (oracle training on real data). This diagnostic is cheap (one regression on held-out Phase 1b data) and converts Theorem 5 into a pre-deployment check.

7 Discussion

What makes the method work. The three ingredients — residual correction, Q^{Δ} filtering, and LLM-guided symbolic features — attack orthogonal failure modes. Without the residual, naive transfer loses the physical-parameter shift. Without the Q^{Δ} filter, extrapolation in the residual is silently accepted. Without the LLM's feature proposals, SINDy's fixed library misses envelope interactions that a domain expert would include.

Relation to formal results. The engineering choice of $\gamma_{\Delta} = 0$ per-step (not Bellman-propagated) Q^{Δ} is the correct choice given Theorem 5: since the residual error $\varepsilon(s, a)$ does not admit a value-style Bellman fixed point on the whole state space, we should weight each transition by its *local* model

accuracy, which is exactly the $\gamma_\Delta = 0$ target in Eq. (2). We verified empirically that $\gamma_\Delta = 0.99$ collapses to a near-constant weight.

Limitations. (i) **Applicability window.** Our method is useful *only* when the sim-real gap is at the physical-dynamics level; we document two cases where it hurts (carpet, ant-wall-broken) and provide a diagnostic threshold for practitioners (Section 6.5). (ii) **LLM dependency.** The LLM oracle is a runtime dependency adding cost ($\approx \$0.17$/run of Claude Haiku); the method could in principle run with a local open model, but the latency–quality trade-off is untested at this time. (iii) **Embodiment diversity.** Our experiments use MuJoCo locomotion (HalfCheetah, Ant, Walker2D). Extending to manipulation benchmarks (e.g., Meta-World, RoboSuite) is future work. (iv) **Assumption gap for Theorem 8.** The rank-equivalence result requires a Gaussian-residual assumption that is violated when the real dynamics are highly non-Gaussian (e.g., contact discontinuities). Role #4's online Kendall-τ diagnostic partially mitigates this by flagging disagreement regions. (v) **Compute-vs-data trade-off.** Our pipeline assumes an accurate simulator is available; when it is not, the MBPO-with-oracle baseline (c4) is preferable at the cost of full real-env training.

Broader Impact. Safe sim-to-real transfer reduces the need for direct real-system training in safety-critical robotics (e.g., autonomous driving, surgical robots), lowering both hardware cost and injury risk. A negative consequence is that our method inherits the LLM provider's biases and failure modes through the feature/constraint proposals (Roles #1, #2); we recommend auditing accepted proposals for any safety-critical deployment. The Pareto gate (Section 4.3) is one such audit mechanism. We report our method's *failure modes* (Section 6.5 and Appendix D) as part of the main text rather than in an afterthought to discourage silent misuse.

8 Conclusion

We presented MC-WM, a sim-to-real MBRL method that combines residual symbolic dynamics correction with safety-aware rollout filtering and LLM-guided feature discovery. On three physics-gap benchmarks we achieve 9.6–$13\times$ the return of naive sim-transfer and match an oracle baseline while using only a modest real-data budget. A theoretical analysis (policy-conditional Residual Simulation Lemma and its universal-convergence counter-example) formally justifies the method's design and predicts its applicability window, which the experiments in Section 6.5 confirm. An ablation identifies the LLM's feature proposals and the Bellman Q^Δ critic as the two most impactful components. All theorems are mechanized in Lean 4 with zero `sorry`.

References

[1] Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. Constrained policy optimization. In *ICML*, 2017.

[2] Philip J Ball et al. Efficient online reinforcement learning with offline data. In *ICML*, 2023.

[3] Stefan Banach. Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3:133–181, 1922.

[4] Steven L Brunton, Joshua L Proctor, and J Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *PNAS*, 2016.

[5] Xinyue Chen, Che Wang, Zijian Zhou, and Keith Ross. Randomized ensembled double q-learning: Learning fast without a model. In *ICLR*, 2021.

[6] Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *NeurIPS*, 2018.

[7] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *CADE*, 2021.

[8] Yuqing Du et al. Guiding pretraining in reinforcement learning with large language models. In *ICML*, 2023.

[9] Benjamin Eysenbach et al. Off-dynamics reinforcement learning: Training for transfer with domain classifiers. In *ICLR*, 2021.

[10] Danijar Hafner et al. Dream to control: Learning behaviors by latent imagination. In *ICLR*, 2020.

[11] Nicklas Hansen, Xiaolong Wang, and Hao Su. TD-MPC: Temporal difference learning for model predictive control. In *ICML*, 2022.

[12] Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *NeurIPS*, 2019.

[13] Michael Kearns and Satinder Singh. Near-optimal reinforcement learning in polynomial time. *Machine Learning*, 49:209–232, 2002.

[14] Ashish Kumar, Zipeng Fu, Deepak Pathak, and Jitendra Malik. RMA: Rapid motor adaptation for legged robots. *arXiv preprint arXiv:2107.04034*, 2021.

[15] John B. Lanier, Micah Baker, Kaiqu Xu, Minghao Liu, Zhaojin Yang, Rahul Bhattacharya, Alan Fern, and Yao-Hung Hubert Tsai. Adapting world models with latent-state dynamics residuals. In *ICRA Workshop*, 2024.

[16] Kimin Lee et al. Sunrise: A simple unified framework for ensemble learning in deep reinforcement learning. In *ICML*, 2021.

[17] Yecheng Jason Ma et al. Eureka: Human-level reward design via coding large language models. *arXiv preprint arXiv:2310.12931*, 2023.

[18] Shehryar Malik, Usman Anwar, Alireza Aghasi, and Ali Ahmed. Inverse constrained reinforcement learning. In *ICML*, 2021.

[19] Haoyi Niu, Tianying Ji, Bingqi Liu, Haocheng Zhao, Xiangyu Zhu, Jianying Zheng, Pengfei Huang, Xianyuan Zhan, Jianming Zhu, and Jun Hu. Beyond OOD state actions: Supported cross-domain offline reinforcement learning. In *AAAI*, 2024.

[20] Xue Bin Peng et al. Sim-to-real transfer of robotic control with dynamics randomization. In *ICRA*, 2018.

[21] Alex Ray, Joshua Achiam, and Dario Amodei. Benchmarking safe exploration in deep reinforcement learning, 2019.

[22] Stéphane Ross and J. Andrew Bagnell. Agnostic system identification for model-based reinforcement learning. In *ICML*, 2012.

[23] Russ Tedrake and Scott Kuindersma. Bridging the sim-to-real gap with dynamic compliance control. In *ICRA Workshop on Robotic Manipulation*, 2023.

[24] Josh Tobin et al. Domain randomization for transferring deep neural networks from simulation to the real world. In *IROS*, 2017.

[25] Andrew Trask, Felix Hill, Scott E Reed, Jack Rae, Chris Dyer, and Phil Blunsom. Neural arithmetic logic units. In *NeurIPS*, 2018.

A Implementation Details

Compute. All experiments on a single NVIDIA RTX 4070 Super (8 GB VRAM). A full c9 run on `gravity_soft_ceiling` takes approximately 45–55 minutes wall-clock, of which ~ 20 minutes is Phase 1b (SINDy feature expansion + NAU training) and the remainder Phase 2 (MBPO). Memory footprint per process is ≈ 1.5 GB VRAM and ≈ 2.5 GB RAM. With four parallel runs sharing the same GPU we observed $\sim 3.3\times$ slowdown per run, suggesting a Pareto-optimal choice of two parallel runs on this hardware.

Hyperparameters. *SAC policy.* LCB critic with pessimism $\beta = 1.0$; $N = 5$ critics; policy MLP $[256, 256]$ with tanh squash; learning rate 3×10^{-4} with Adam; target smoothing $\tau = 0.005$; batch 256; discount $\gamma = 0.99$. *World model.* Ensemble of $K = 5$ probabilistic networks with $[200, 200]$ hidden; predicts $(\Delta s, r)$; trained for 100 epochs with patience 20 on 50k sim transitions. *Residual* $\hat{\delta}$. SINDy library starts from polynomial degree 2 (bias + linear + pairwise) plus a sine/cosine block on select state dims; STLSQ threshold 0.05 calibrated from median residual MSE; up to $R_{\max} = 3$ LLM feature-expansion rounds. NAU head: MLP $[64, 64]$ on the SINDy feature vector, 100 epochs with Lipschitz penalty $\lambda_L = 0.01$ aiming for $L_{\text{eff}} \leq 100$. Q^{Δ} *critic.* MLP $[128, 128]$ with ReLU; τ initialized from median $\|\hat{M}(s, a) - s'\|^2$ on a 500-sample batch; $\gamma_{\Delta} = 0.5$ (ablated in Section 6.4 at $\gamma_{\Delta} = 0$). *MBPO rollouts.* Horizon 1; rollout batch 400; rollout freq every $k_{\text{roll}} = 1000$ env steps. Model refit every $k_{\text{refit}} = 1000$ env steps on the most recent 50k transitions (50 epochs, patience 15). Constraint rejection threshold $w < 0.01$. *LLM cadence.* Role #1 once in Phase 1b and when Role #3 fires; Role #2 up to 3 rounds in Phase 1b; Role #3 audit every 10,000 env steps (top-$p = 95\%$); Role #5 every 5000 env steps. Pareto violation cap $v_{\text{cap}} = 20$/episode. All seeds $\in \{42, 123, 456\}$.

LLM cost. Claude Haiku 4.5 API at \$1.00 per million input tokens and \$5.00 per million output tokens (as of April 2026). Per-run token budget: Role #1 ($\sim$ 4k in / 1k out $\times$3 calls), Role #2 ($\sim$ 6k in / 2k out $\times$3 calls), Role #3 ($\sim$ 5k / 1k $\times$5 calls), Role #5 ($\sim$ 2k / 0.5k $\times$10 calls). Total $\sim$ 85k input + 17k output tokens per run, i.e. approximately \$0.17 per run. Compared with the $\sim$ 45-minute wall-clock cost of GPU compute this is negligible.

Reproducibility. Code: `https://anonymous.4open.science/r/MC-WM-<redacted>`. Environment specifications (MuJoCo XML modifications for the gravity/ friction shifts) are in `mc_wm/envs/hp_mujoco/`. Seeds are set globally via `numpy.random.seed`, `torch.manual_seed`, and the Gym env seed. Non-determinism from Claude outputs is the only source of seed-independent variance; we report full per-seed tables in Appendix B to allow auditing.

B Full Per-Seed Tables

Table 6: Per-seed results. Values are the last-3-evaluation mean of eval return and mean violations/episode.

gravity_ceiling	s42	s123	s456	mean
c1 return	713	659	256	543
c1 viol	0.53	0.70	0.93	0.72
c9 return	5374	5290	4925	5196
c9 viol	0.00	0.00	0.07	0.02
gravity_soft_ceiling	s42	s123	s456	mean
c1 return	131	726	428	428
c1 viol	18.2	6.8	8.8	11.3
c4 return	5605	6507	5983	6032
c4 viol	0.30	1.27	0.00	0.52
c9 return	6611	5554	6938	6368
c9 viol	1.4	2.4	5.9	3.2
H2O+ return	2276	2448	1572	2099
friction_walker_soft_ceiling	s42	s123	s456	mean
c1 return	266	230	291	262
c1 viol	8.6	6.6	0.3	5.2
c4 return	290	266	206	254
c4 viol	34.0	5.9	10.0	16.6
c9 return	425	301	345	357
c9 viol	2.1	4.0	20.7	8.9

C LLM Prompts

All role prompts are packaged in the supplementary material (`mc_wm/oracle/prompts/`). We summarize each role's input/output structure here. The prompts are few-shot: each begins with an environment-specific system message, followed by 1–3 in-context examples, then the actual query.

Role #1 (Constraint proposer). *Input fields:* environment description; current constraint set $\mathcal{C}$; a sample of 20 recent rejected rollouts for context. *Output format:* JSON list of $\{\text{name}, \text{predicate_python_expr}, \text{rationale}\}$. Example output entry:

> *{"name": "torso_below_ground", "predicate": "s[0] < 0.0", "rationale": "If the torso z-coordinate is negative, the agent has fallen through the ground plane (physical impossibility). Reject any rollout where the next-state satisfies this."}*

Constraints are validated against the real-data buffer: accept iff false-positive rate $<1\%$ on real transitions.

Role #2 (Feature proposer). *Input fields:* per-dimension residual diagnostics (heteroscedasticity metric ranked by "culprit"; excess kurtosis; non-stationary drift); list of N already-accepted features (names only, to avoid over-long context); environment description. *Output format:* JSON list of $\{\text{name}, \text{python_expr}, \text{std_estimate}\}$. The Python expression is `eval`-ed against sample inputs to compute the feature column; standard deviation is reported so the acceptance test can normalize across features. Example output entry:

> *{"name": "tanh_vel_19", "expr": "np.tanh(s[19])", "std": 0.804}*

Role #3 (Audit). *Input fields:* a sample of 50 post-constraint-filter transitions with the largest $\|\hat{\delta}\|$ (top p-percentile); the current constraint set; recent reward and violation trends. *Output format:* JSON list specifying (i) new constraints to propose (format of Role #1 output) and (ii) a decision on whether any existing constraints are too strict (flagged by false-rejection rate). The Pareto gate (Section 4.3) filters the output to prevent reward-seeking proposals during violation spikes.

Role #4 (ICRL φ^* comparator, diagnostic). *Input fields:* a batch of 500 rollout transitions with $Q^{\Delta}(s, a)$ scores and $\varphi(s, a)$ scores (the latter trained online). *Output format:* Kendall-τ correlation between the two rankings, plus a list of the 10 transitions with the largest disagreement (for inspection).

Role #5 (HP orchestrator). *Input fields:* recent reward and violation history (last 5 evaluations); current $(\text{rollout_batch}, \gamma_{\Delta}, \text{audit_percentile})$; a two-entry context summarizing previous HP changes. *Output format:* JSON proposed updates, each annotated with the metric expected to improve. The Pareto gate rejects any reward-seeking proposal (raises to `rollout_batch` or γ_{Δ}) when 3-eval-mean violation exceeds $v_{\text{cap}} = 20$/episode.

D Additional Detail on Negative Results

The headline numbers for the two negative-result environments appear in Table 5 of the main paper (Section 6.5). Both environments have small or no physical-dynamics gap (velocity observation scaling; disabled actuators), so the residual $\hat{\delta}$ fits measurement noise rather than a genuine dynamics term. The resulting imagined transitions are biased enough that the Q^{Δ} filter cannot recover naive-transfer performance, no matter how aggressively we down-weight the imagined batch.

Appendix E reports a β-scaling mitigation (a learned scalar on the residual contribution) that also fails to close the gap, consistent with the structural reading of Thm. 5: the problem is not that $\hat{\delta}$ is too large in magnitude, it is that $\hat{\delta}$'s *direction* encodes noise rather than dynamics.

E β-Scaling Adaptation Experiments (Failed)

We attempted to mitigate the negative results of Appendix D by scaling the residual's contribution through a learned coefficient β (BAPR-HRO-style bandit over β; NAU-improvement-informed β;

continuous EXP3 / hill-climbing / multi-β rollouts; and per-state ensemble-gated β). Ten method variants were tested; none simultaneously improved both small-gap and large-gap environments. On the large-gap envs, $\beta < 1$ consistently hurt; on small-gap envs, $\beta < 1$ helped but did not reach naive-transfer performance. We conclude that the failure mode is not a miscalibrated scalar but the structural issue of Theorem 5. Details are in the supplementary material.

F Proofs

We formalize Theorems 3–6 in Lean 4 [7]. The file `ResidualMDP.lean` compiles under Mathlib 4 with zero `sorry` and only three axioms that we state explicitly below. Key definitions:

- `FiniteMDP` – finite-state MDP with stochastic kernel, bounded reward, and $\gamma < 1$.
- `ResidualKernel M_sim` – a kernel $\hat{\delta}$ such that $P_{\text{sim}} + \hat{\delta}$ is stochastic.
- `bellmanOp`, `V_pi` – Bellman policy operator and its fixed-point value function.
- `visitationSupport M pol` – $\text{supp}(d^{\pi}_{\mathcal{M}})$.
- `kernelTV_sup P Q U` – TV supremum restricted to U.

F.1 Full proof of Theorem 3

Let $e(s) := V^{\pi}_{\mathcal{M}_{\text{real}}}(s) - V^{\pi}_{\mathcal{M}_{\text{resid}}}(s)$ and $\varepsilon := \text{TV}_{\text{supp}(d^{\pi}_{\mathcal{M}_{\text{real}}})}(P_{\text{real}}, P_{\text{resid}})$. The proof proceeds in six steps.

Step 1 (triangle-decomposition). For every s, $V^{\pi}_{\mathcal{M}}(s) = (T^{\pi}_{\mathcal{M}} V^{\pi}_{\mathcal{M}})(s)$ is the Bellman fixed-point identity. Using this, $V^{\pi}_{\mathcal{M}_{\text{real}}}(s) - V^{\pi}_{\mathcal{M}_{\text{resid}}}(s) = (T^{\pi}_{\mathcal{M}_{\text{real}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s) - (T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{resid}}})(s)$. Introduce the intermediate term $(T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s)$ and triangle-inequality:

$$|e(s)| \le \underbrace{\left|(T^{\pi}_{\mathcal{M}_{\text{real}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s) - (T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s)\right|}_{B_1(s)} + \underbrace{\left|(T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{real}}})(s) - (T^{\pi}_{\mathcal{M}_{\text{resid}}} V^{\pi}_{\mathcal{M}_{\text{resid}}})(s)\right|}_{B_2(s)}.$$

Step 2 ($\sup$ **of sum** $\le$ **sum of** $\sup$**).** Taking supremum over $s \in \mathcal{S}$ (finite) and using sub-additivity of $\sup$ for finite index sets, $\|e\|_\infty \le \sup_s B_1(s) + \sup_s B_2(s)$.

Step 3 (bracket B_1 — TV + Hölder bound). $B_1(s)$ applies two different kernels to the *same* $V = V^{\pi}_{\mathcal{M}_{\text{real}}}$:

$$B_1(s) = \left|\gamma \sum_a \pi(a|s) \sum_{s'} [P_{\text{real}}(s'|s,a) - P_{\text{resid}}(s'|s,a)] V(s')\right|.$$

Pointwise Hölder: for any probability measures P, Q and bounded V, $|E_P V - E_Q V| \le 2\|V\|_\infty \cdot \text{TV}(P, Q)$. Combining with $\|V\|_\infty = \|V^{\pi}_{\mathcal{M}_{\text{real}}}\|_\infty \le R_{\max}/(1-\gamma)$ and restricting the TV supremum to (s, a) pairs in $\text{supp}(d^{\pi}_{\mathcal{M}_{\text{real}}})$ (terms outside the support contribute 0 to the discounted return expectation, see Step 3b of the Lean proof for the formal argument):

$$\sup_s B_1(s) \;\le\; \gamma \cdot \frac{2R_{\max}}{1-\gamma} \cdot \varepsilon.$$

Step 4 (bracket B_2 — Bellman contraction). $B_2(s)$ applies the *same* operator $T^{\pi}_{\mathcal{M}_{\text{resid}}}$ to two different value functions; by Lemma 9, $\sup_s B_2(s) \le \gamma \|V^{\pi}_{\mathcal{M}_{\text{real}}} - V^{\pi}_{\mathcal{M}_{\text{resid}}}\|_\infty = \gamma \|e\|_\infty$.

Step 5 (combine). From Steps 2–4: $\|e\|_\infty \le \frac{2\gamma R_{\max}}{1-\gamma}\varepsilon + \gamma\|e\|_\infty$.

Step 6 (solve). Rearranging, $(1-\gamma)\|e\|_\infty \le \frac{2\gamma R_{\max}}{1-\gamma}\varepsilon$, hence $\|e\|_\infty \le \frac{2\gamma R_{\max}}{(1-\gamma)^2}\varepsilon$. □

F.2 Full proof of Theorem 5

Fix $\gamma = 0.5$. Construct a 2-state 1-action MDP $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a\}$, reward $r(s_1, a) = 0$, $r(s_2, a) = 1$. Set $P_{\text{sim}}(s_1|s_1, a) = P_{\text{sim}}(s_2|s_2, a) = 1$ (self-loops), and $P_{\text{real}}(s_1|s_1, a) = 1$, $P_{\text{real}}(s_1|s_2, a) = 1$, $P_{\text{real}}(s_2|s_2, a) = 0$ (all real trajectories eventually absorb in s_1). Let $\pi_{\text{data}}(a|s_1) = 1$, $\pi_{\text{data}}(a|s_2) = 0$; then $d^{\pi}_{\pi_{\text{data}}}$ concentrates on s_1, so $\text{supp}(d^{\pi}_{\pi_{\text{data}}}) = \{(s_1, a)\}$. Any $\hat{\delta}$ with $\hat{\delta}(\cdot|s_1, a) = 0$ satisfies $\text{TV}_{\text{supp}(d^{\pi}_{\pi_{\text{data}}})}(P_{\text{real}}, P_{\text{sim}} + \hat{\delta}) = 0$; we choose $\hat{\delta}(s_2|s_2, a) = 1$ so that $\mathcal{M}_{\text{resid}}$ has s_2 as an absorbing state (the same as P_{sim}). Now consider $\pi'(a|s_2) = 1$. Then $V^{\pi'}_{\mathcal{M}_{\text{real}}}(s_2) = 0$ (escapes to s_1 next step, reward 0 forever after) but $V^{\pi'}_{\mathcal{M}_{\text{resid}}}(s_2) = \sum_{t\geq 0} \gamma^t \cdot 1 = 1/(1-\gamma) = 2$. Hence $\|V^{\pi'}_{\mathcal{M}_{\text{real}}} - V^{\pi'}_{\mathcal{M}_{\text{resid}}}\|_\infty \geq 2 > 0$ despite perfect fit on π_{data}'s support. □

F.3 Full proof of Theorem 8

Under Assumption 2, for the Bayes-optimal discriminator φ^* on data pairs $\{(s, a) \mapsto \text{real}, (s, a) \mapsto \text{sim}\}$ we have $\log \frac{\varphi^*(s,a)}{1-\varphi^*(s,a)} = \log \frac{p_{\text{real}}(s'|s,a)}{p_{\text{sim}}(s'|s,a)}$. Substituting two Gaussians of the same variance σ^2, $\log p_{\text{real}} - \log p_{\text{sim}} = (e_{\text{sim}} - e_{\text{real}})/(2\sigma^2)$ where $e_X(s, a) := \|\mu_X(s, a) - s'\|^2$. Rewriting $e_{\text{sim}} = e_{\text{resid}} + \text{const}$ (since $\mathcal{M}_{\text{resid}}$ is a valid kernel close to P_{sim} on support), we obtain the log-odds formula quoted in the main text. The Q^Δ log-target $\log Q^\Delta = -e_{\text{real}}/\tau$ ($\tau = 2\sigma^2$) is direct from the squared-residual definition in Eq. (2) with $\gamma_\Delta = 0$. The difference $\log \frac{\varphi^*}{1-\varphi^*} - \log Q^\Delta = e_{\text{resid}}/\tau$ is a function of e_{resid} alone. Any constant cannot change rankings; a δ-bounded variation can only swap pairs whose $|\log Q^\Delta|$-gap is at most $2\delta/\tau$. □

F.4 Full proof of Theorem 6

The refit operator F factors as the composition $F = F_{\text{reg}} \circ F_{\text{visit}} \circ F_{\text{pol}}$ of three maps: $F_{\text{pol}} : \hat{\delta} \mapsto \pi[\hat{\delta}]$ (policy induction), $F_{\text{visit}} : \pi \mapsto d^\pi$ (visitation), $F_{\text{reg}} : (\hat{\delta}, d^\pi) \mapsto \text{argmin}_{\hat{\delta}'} \mathbb{E}_{(s,a)\sim d^\pi} \|(P_{\text{real}} - P_{\text{sim}} - \hat{\delta}')(\cdot|s, a)\|_2^2$ (L^2 regression). By Assumption 1(a)(b)(c), F_{pol}, F_{visit}, and F_{reg} are Lipschitz with constants L_π, L_d, and B, respectively. Composition of Lipschitz maps gives $\|F(\hat{\delta}_1) - F(\hat{\delta}_2)\|_\infty \leq L_\pi L_d B \, \|\hat{\delta}_1 - \hat{\delta}_2\|_\infty$. Since the residual function space is a closed subset of the finite-dimensional Euclidean space $\mathbb{R}^{|\mathcal{S}|^2|\mathcal{A}|}$ (hence complete under sup-norm), Banach's fixed-point theorem [3] applies whenever $L_\pi L_d B < 1$, giving the unique fixed point $\hat{\delta}^*$ and geometric convergence rate $(L_\pi L_d B)^k$. □

Lean axioms. Only three axioms are used in `ResidualMDP.lean`: (A1) measure-theoretic TV-Hölder identity (`bellman_op_kernel_diff`; provable from standard Mathlib but requires additional setup outside our scaffold); (A2) Bayes-optimal discriminator log-odds identity (`phi_bayes_optimal_logodds`); and (A3) Banach fixed-point theorem for finite-dimensional residual space (`banach_finite_residual`; available in Mathlib as `ContractingWith.fixedPoint` but requires metric-space instance boilerplate). We flag these explicitly as `axiom` in the Lean source. Every `theorem` in `ResidualMDP.lean` has zero `sorry`.

We refer the reader to the Lean file (supplementary material) for the mechanized proofs. We highlight here only the key Lemma 9.

Lemma 9 (Bellman γ-contraction)**.** *For any finite MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with $\gamma \in [0, 1)$ and any policy π, $\|T^\pi_{\mathcal{M}} V_1 - T^\pi_{\mathcal{M}} V_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty$.*

Proof. For each state s, $(T^\pi V_i)(s) = \sum_a \pi(a|s)[r(s, a) + \gamma \sum_{s'} P(s'|s, a)V_i(s')]$. The difference of the two operators at s is γ times an expectation of $(V_1 - V_2)$ under a probability measure, and an expectation is bounded above by the supremum, giving $|(T^\pi V_1)(s) - (T^\pi V_2)(s)| \leq \gamma \|V_1 - V_2\|_\infty$. Taking sup over s completes the proof. □